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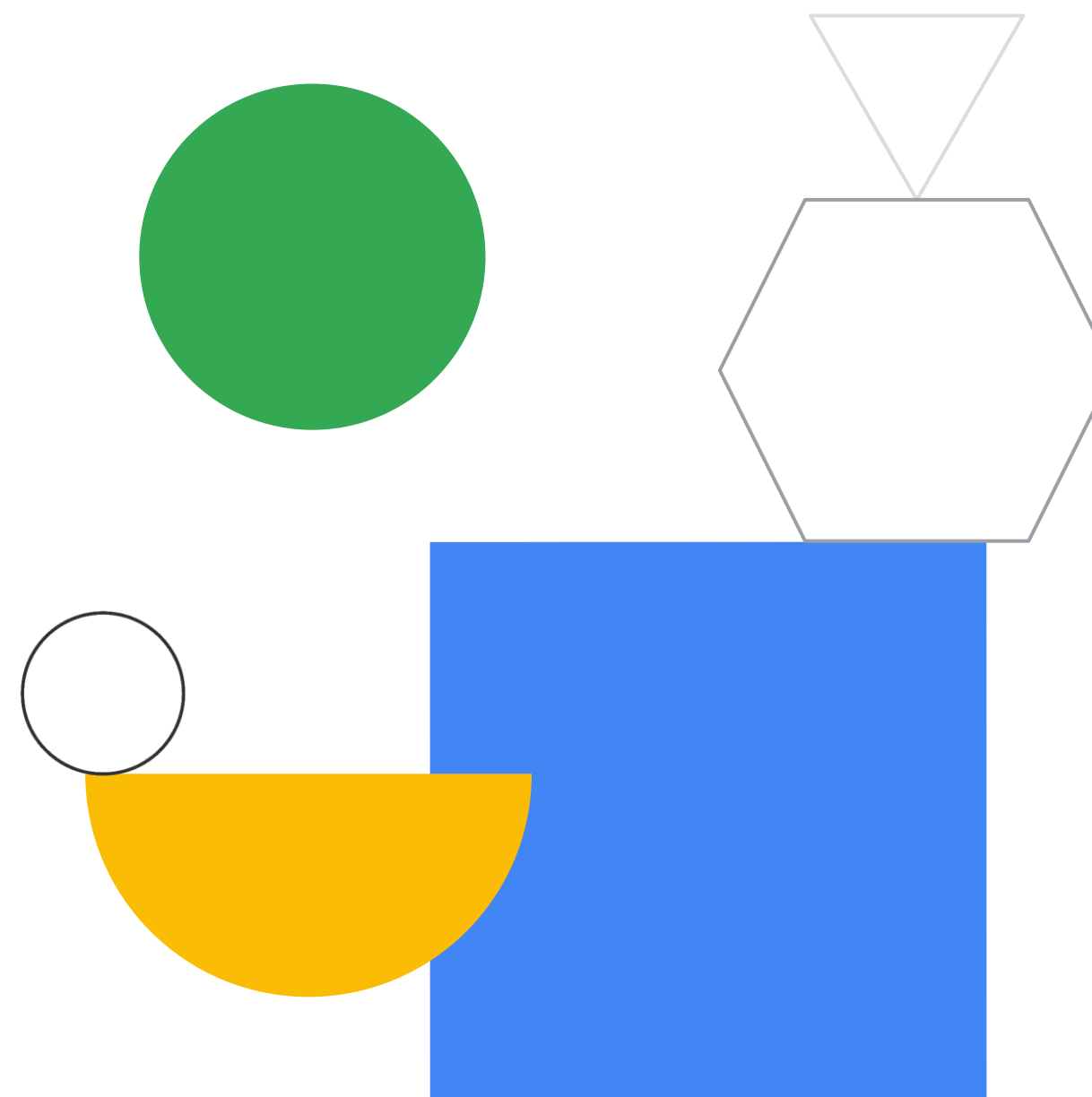
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Use Google Cloud to build your apps

Module 3 - Compute
Google Cloud Computing Foundations



Course map

01 —

So, what's the cloud anyway?

02 —

Start with a solid platform

03 —

Use Google Cloud to build your apps

04 —

Where do I store this stuff?

05 —

There's an API for that!

06 —

You can't secure the cloud, right?

07 —

It helps to network

08 —

Keeping an eye on things

09 —

You have the data, but what are you doing with it?

10 —

Let machines do the work

Google Cloud skill badges

Objectives

- 01 Explore the role of compute options in the cloud.
- 02 Learn about building and managing virtual machines.
- 03 Examine building elastic applications using autoscaling.
- 04 Explore PaaS options by leveraging App Engine.
- 05 Examine building event-driven services using Cloud Functions.
- 06 Identify containerizing and orchestrating applications with Google Kubernetes Engine.
- 07 Identify developing and deploying scalable containerized applications with Cloud Run.





Agenda



- 01 Compute Options in the Cloud
- 02 Exploring IaaS with Compute Engine
- 03 Lab: Creating a Virtual Machine
- 04 Configuring Elastic Apps with Autoscaling
- 05 Exploring PaaS with App Engine
- 06 Lab: App Engine: Qwik Start - Python
- 07 Event Driven Programs with Cloud Functions
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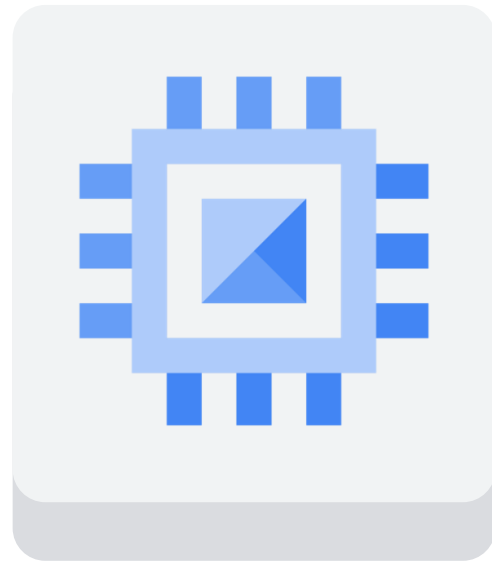


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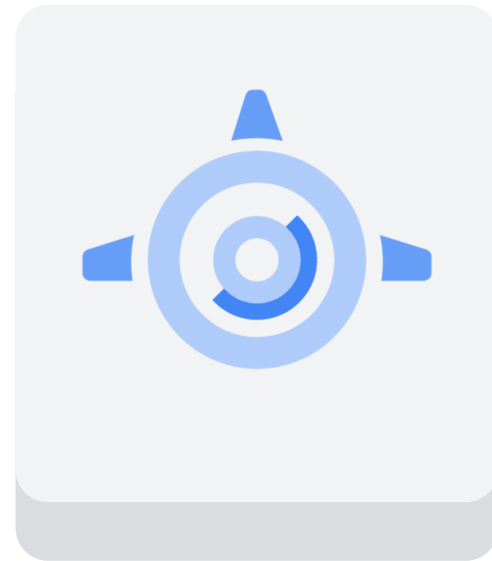
Google Cloud compute services



Compute Engine

IaaS

Virtual machines with industry-leading price/performance



App Engine

PaaS

A flexible, zero ops platform for building highly available apps



Cloud Functions

Serverless logic

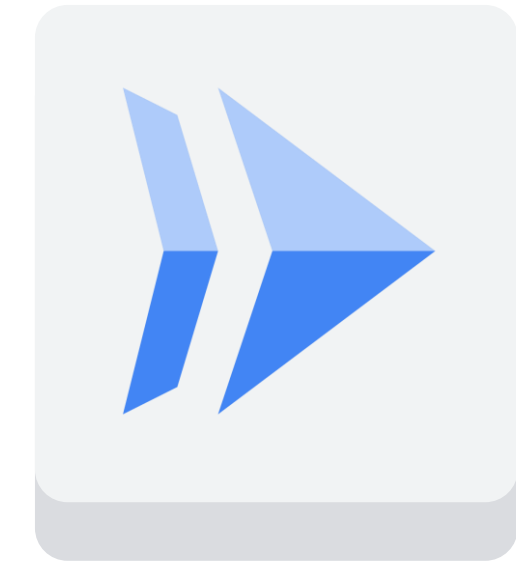
A lightweight fully managed serverless execution environment for building and connecting cloud services



Google Kubernetes Engine

Hybrid

Cluster manager and orchestration engine built on Google's container experience



Cloud Run

Serverless platform

A fully managed serverless platform for developing and deploying scalable containerized applications

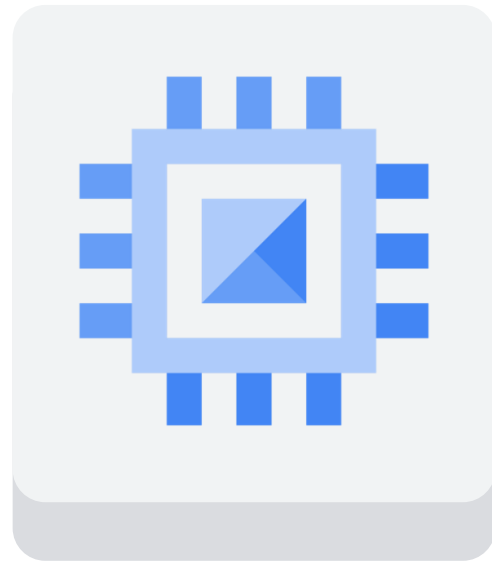


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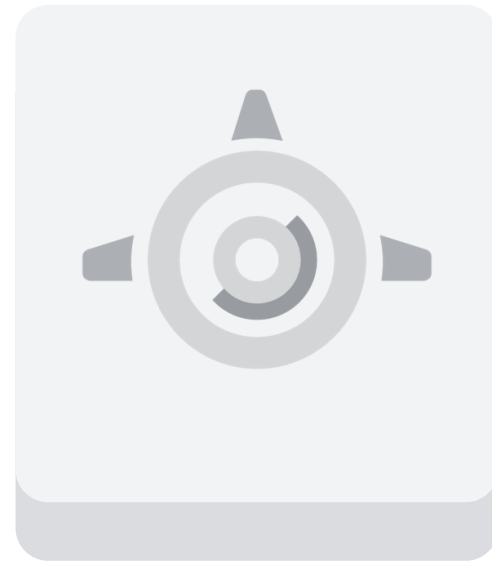
Google Cloud compute services



Compute
Engine

IaaS

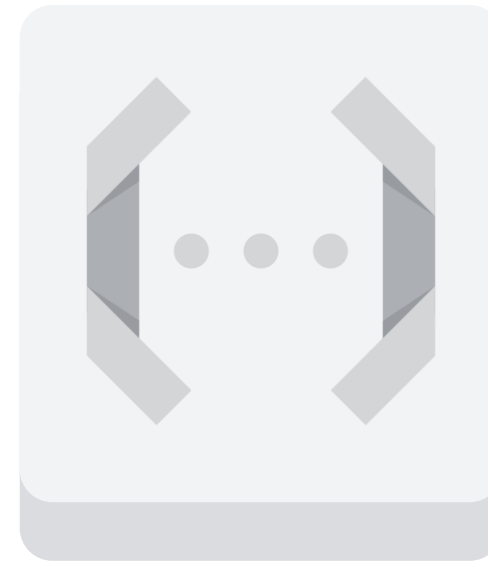
Virtual machines with
industry-leading
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App Engine

PaaS

A flexible, zero ops
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Cloud
Functions

Serverless logic

A lightweight fully
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Google Kubernetes
Engine

Hybrid

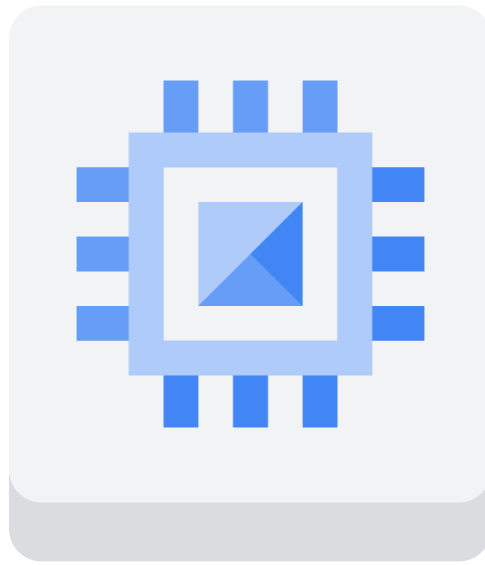
Cluster manager and
orchestration engine
built on Google's
container experience



Cloud Run

Serverless platform

A fully managed
serverless platform for
developing and
deploying scalable
containerized
applications



Compute Engine



No upfront investments



Thousands of virtual CPUs can run on a system that's designed to be fast and offer consistent performance

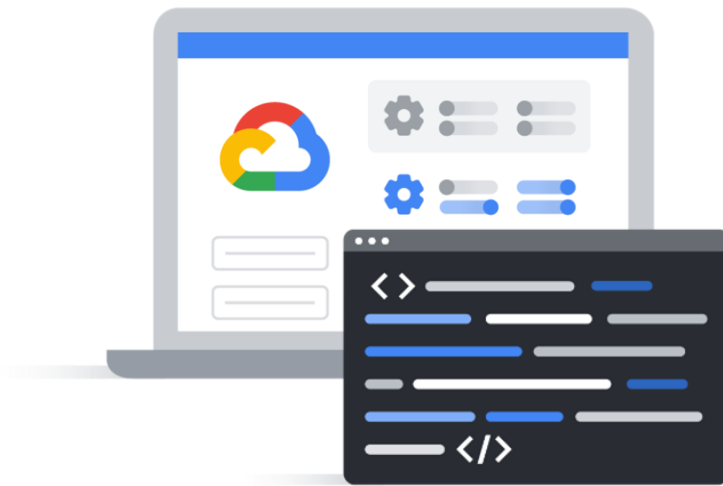


Each VM contains the power and functionality of a full-fledged operating system



Run any computing workload such as web-server hosting, application hosting, and/or application backends

Virtual machine instances are flexible resources



VM can be created via the Google Cloud console or the gcloud command-line tool

Linux Server images

Windows Server images

provided by **Google**

VM can run Linux and Windows Server images or any customized versions of these images



You can build and run images of other operating systems and flexibly reconfigure virtual machines

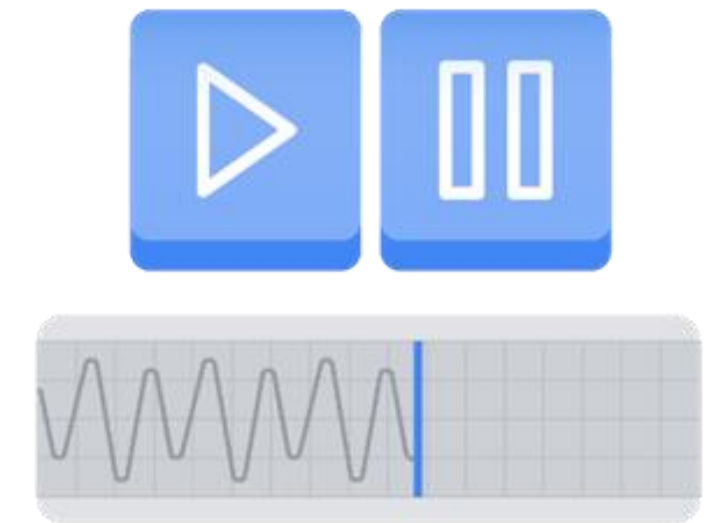
Compute Engine pricing and billing offers discounts



Sustained-use



Committed-use



Preemptible VMs



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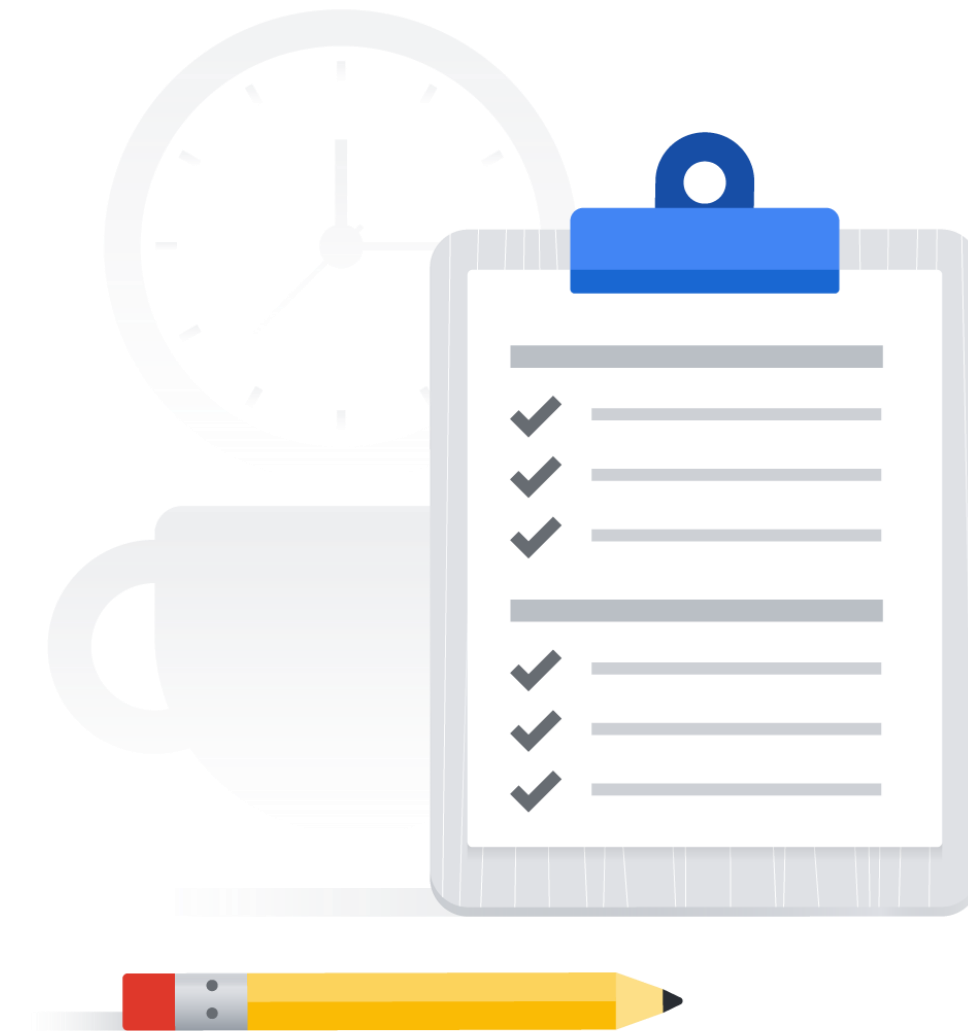
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Lab: Creating a Virtual Machine

 Hands-on lab

During this lab you'll:

1. Create a virtual machine with the Google Cloud console.
2. Create a virtual machine with gcloud command line.
3. Deploy a web server and connect it to a virtual machine.



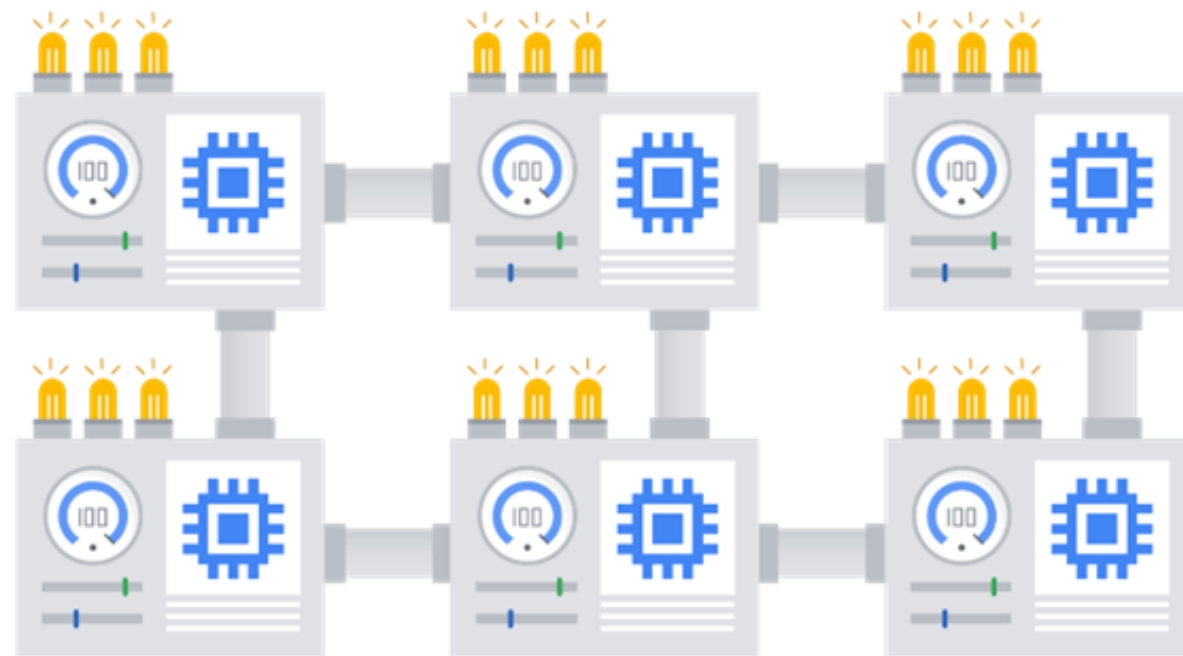


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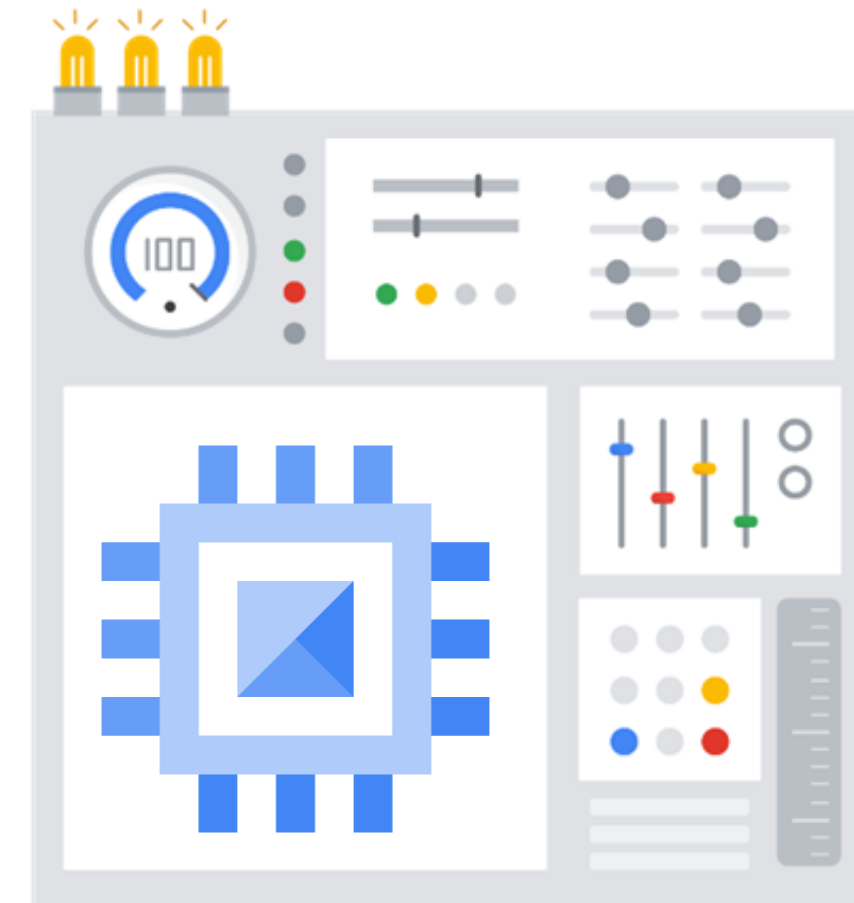


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VMs can be autoscaled to meet demand



Use [autoscaling](#) for resilient, scalable applications



Use [big VMs](#) for memory- and compute-intensive applications

Specs for currently available VM machine types:

cloud.google.com/compute/docs/machine-types

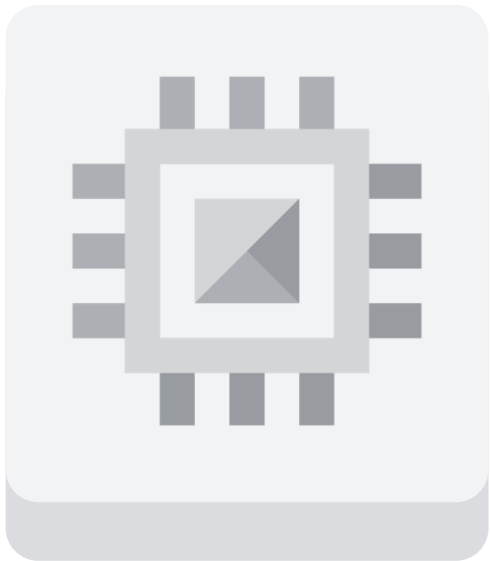


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Google Cloud compute services



Compute Engine



App Engine



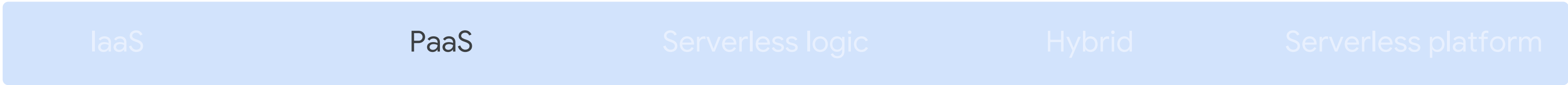
Cloud Functions



Google Kubernetes Engine



Cloud Run



Virtual machines with industry-leading price/performance

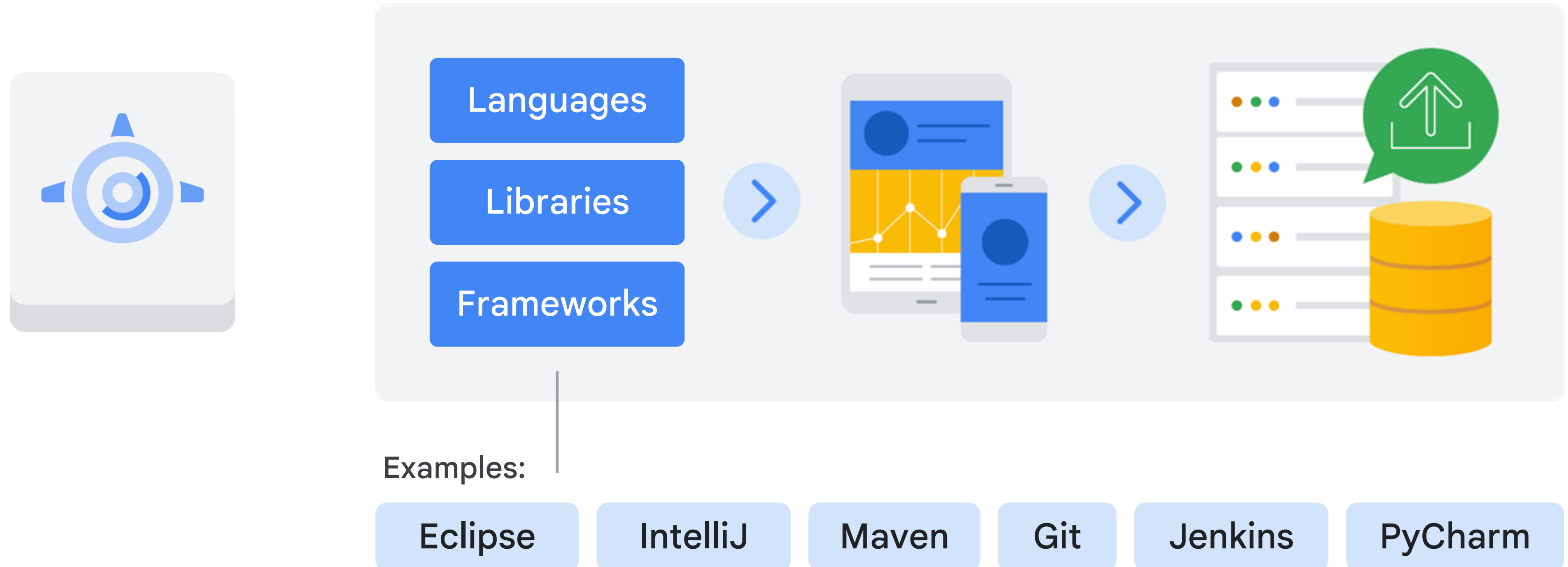
A flexible, zero ops platform for building highly available apps

A lightweight fully managed serverless execution environment for building and connecting cloud services

Cluster manager and orchestration engine built on Google’s container experience

A fully managed serverless platform for developing and deploying scalable containerized applications

App Engine interfaces with many development tools



App Engine provides a full range of built-in services

Built-in services
and APIs



NoSQL datastores

Health checks

Memcache

Application logging

Load balancing

User authentication
API

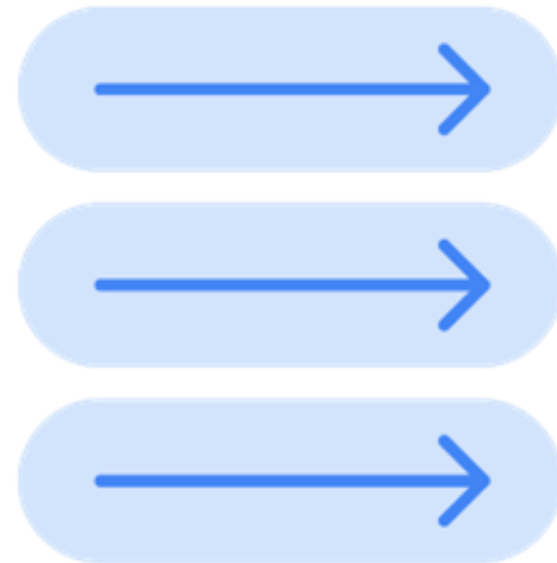
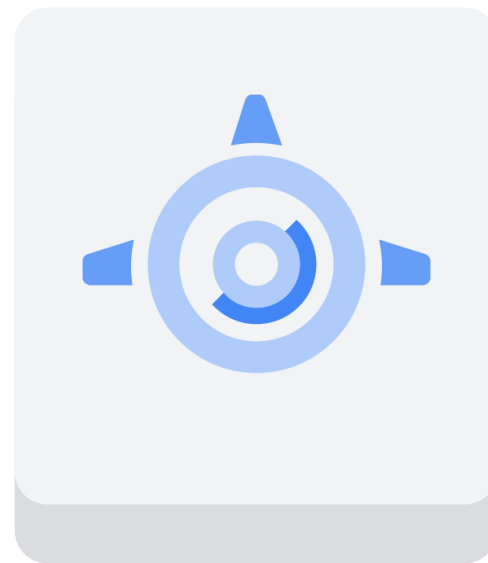
App Engine offers software development kits (SDKs)



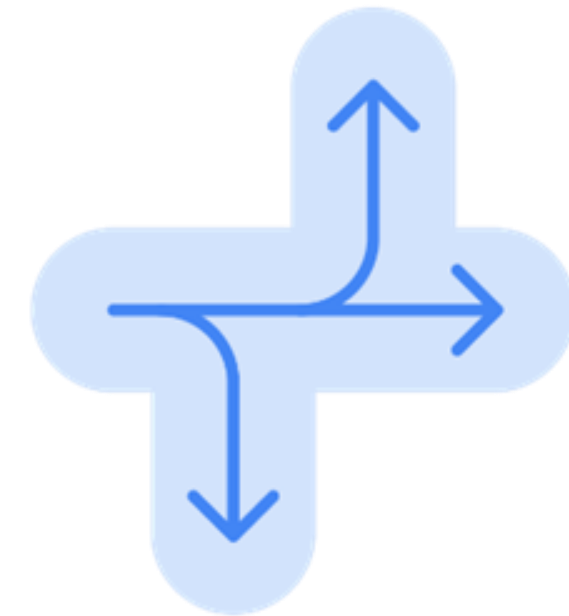
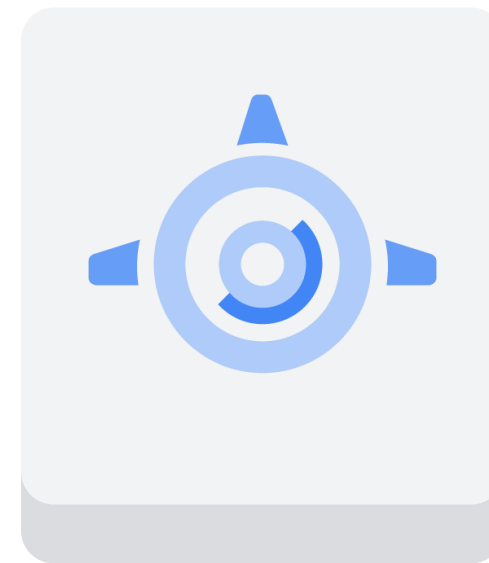
Each SDK includes:

- ✓ APIs and libraries
- ✓ Sandbox environment
- ✓ Deployment tools

Two types of App Engine environments



Standard



Flexible

App Engine standard runs preconfigured containers



Persistent storage with queries, sorting, and transactions

Automatic scaling and load balancing

Asynchronous task queues for performing work outside the scope of a request

Scheduled tasks for triggering events at specified times or regular intervals

Integration with other Google Cloud services and APIs

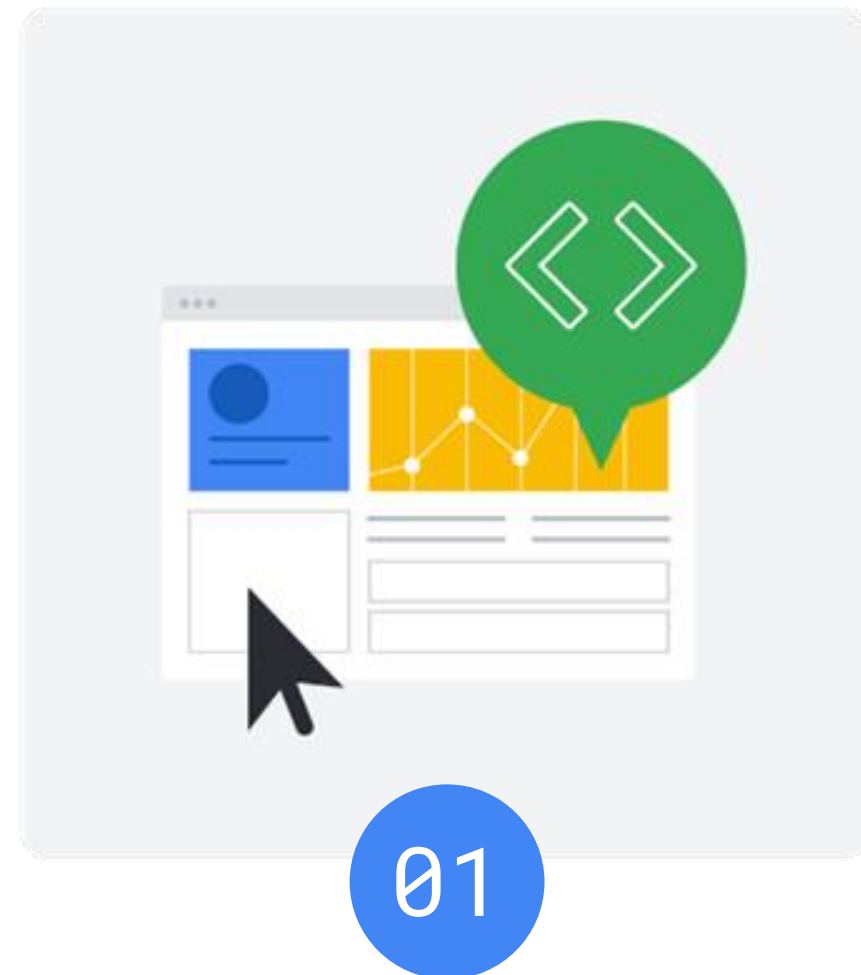
The standard environment has two requirements



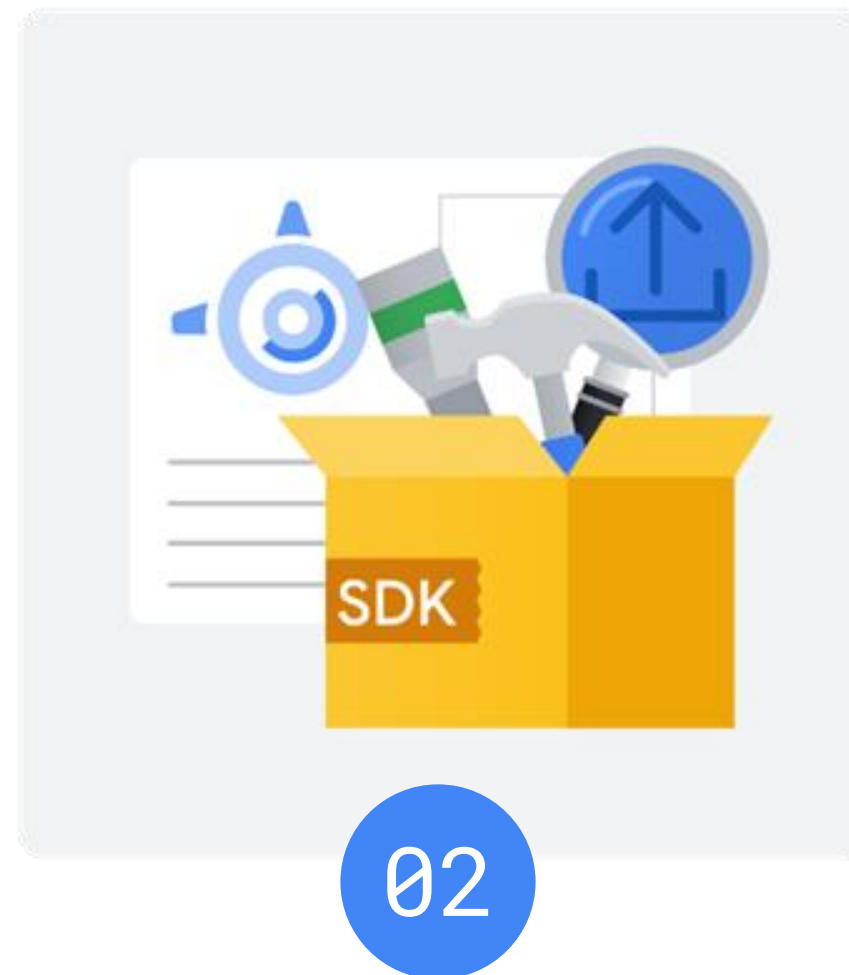
01 Use specified versions of Java, Python, PHP, Go, Node.js, and Ruby

02 Application must conform to sandbox constraints that are dependent on runtime

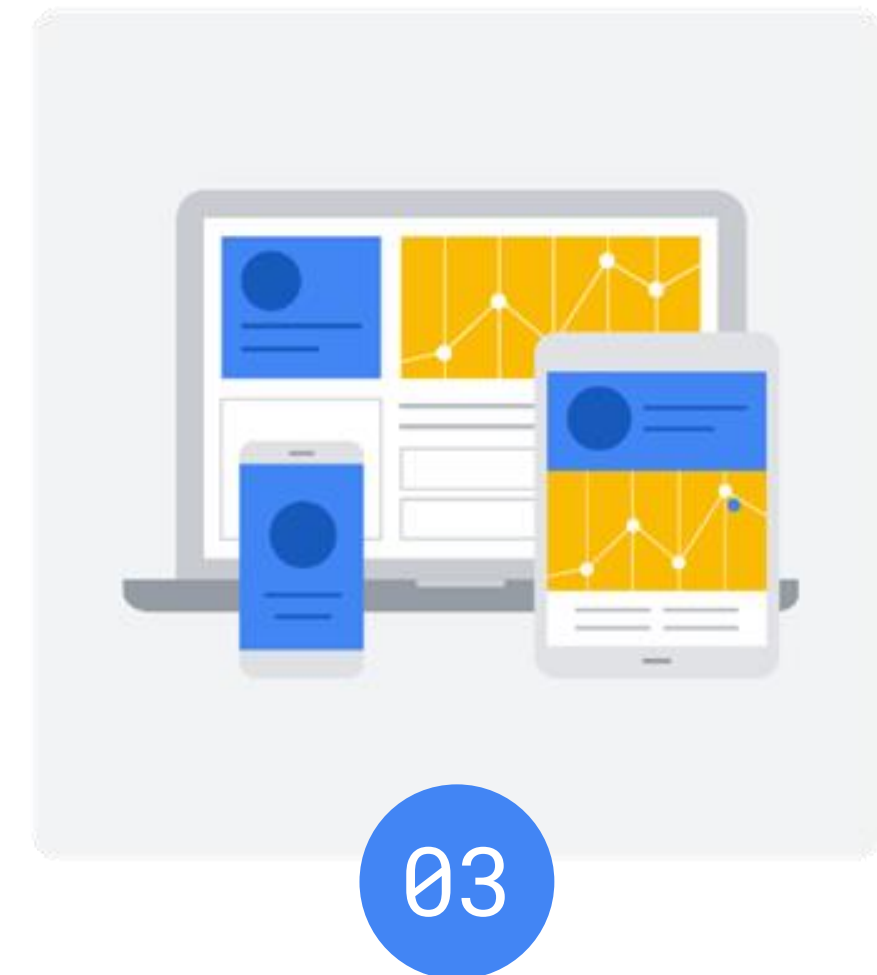
The standard environment workflow



Develop web app
and test locally

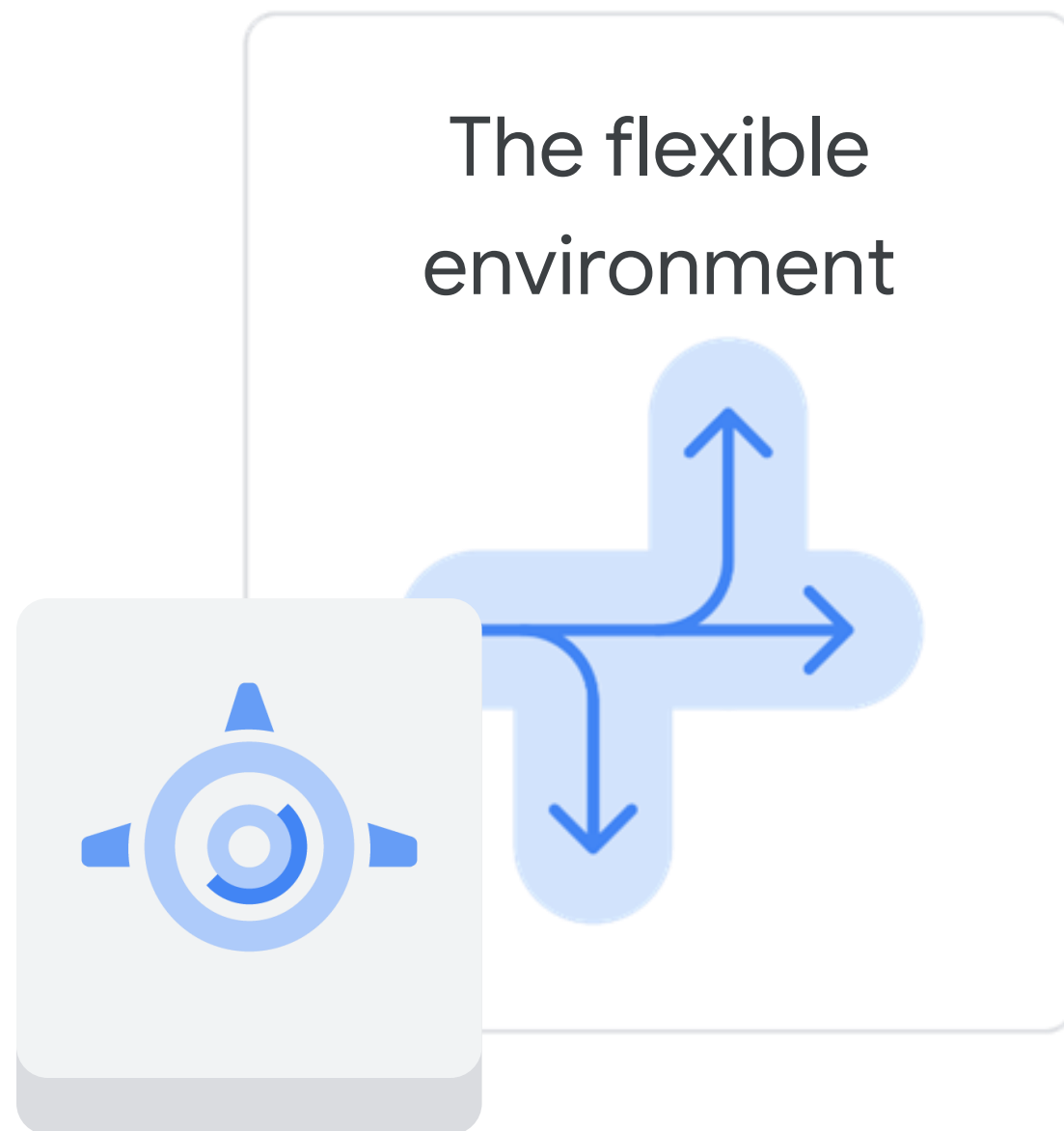


Deploy to App
Engine with SDK



App Engine **scales**
and services the app

App Engine flexible environment uses custom containers



Instances are health-checked, healed, and co-located

Critical, backward-compatible updates are automatically applied to the underlying operating system

VM instances are automatically located by geographical region according to the settings in your project

VM instances are restarted on a weekly basis

The flexible environment features

✓ Microservices

✓ Authorization

✓ SQL & NoSQL databases

✓ Traffic splitting

✓ Logging

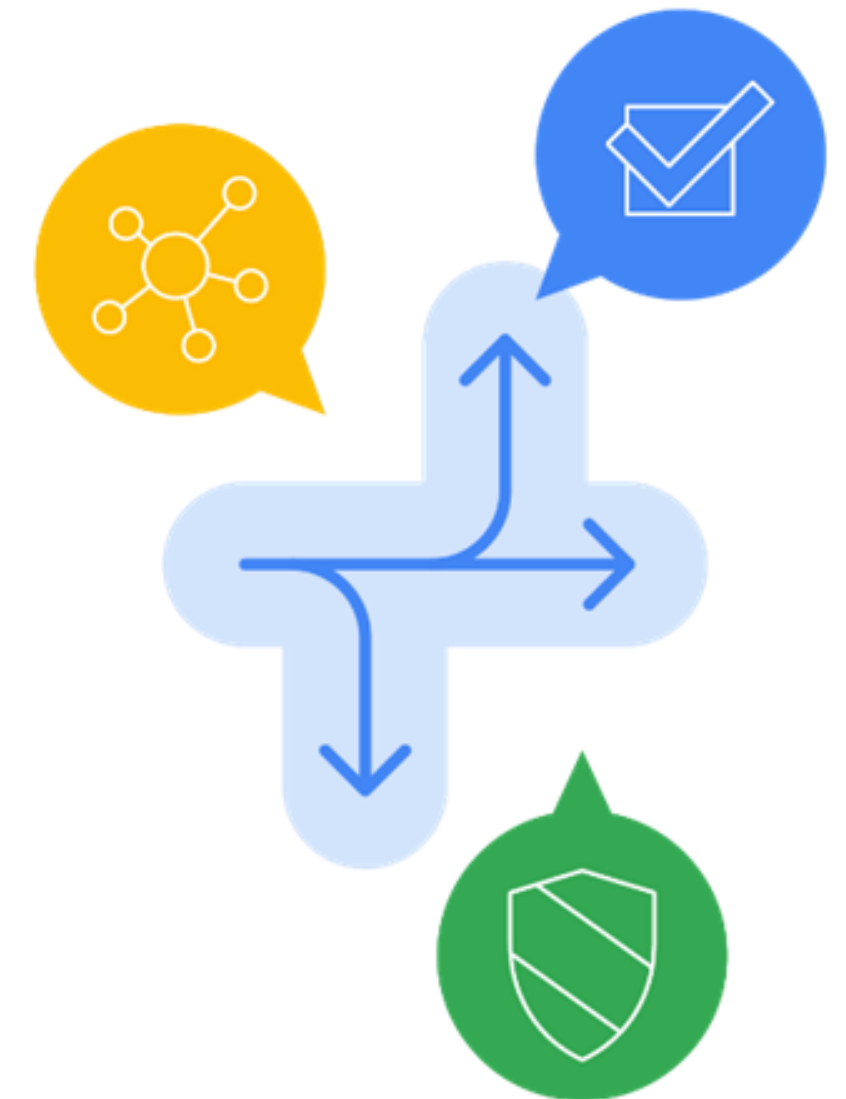
✓ Search

✓ Versioning

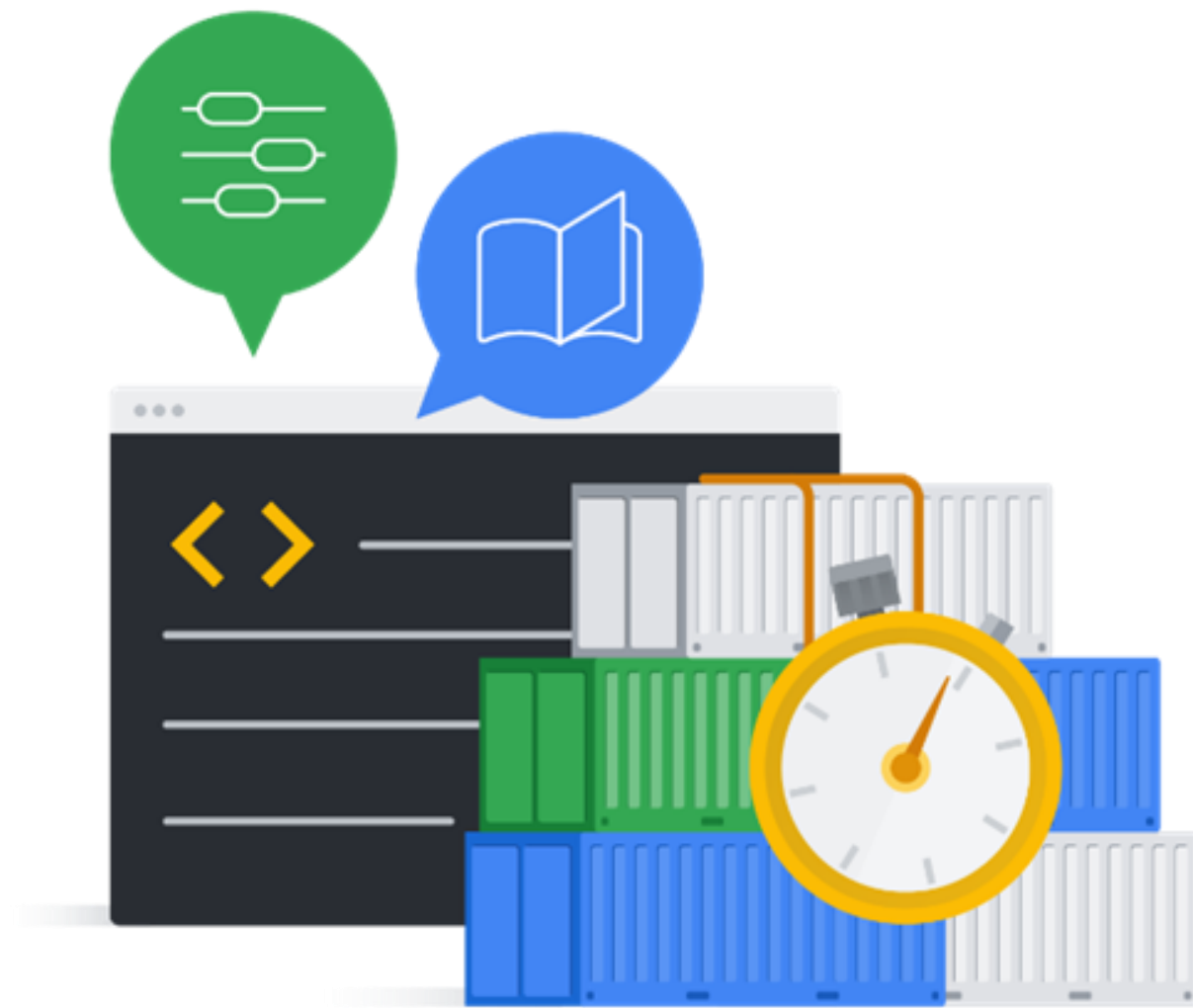
✓ Security scanning

✓ Memcache

✓ Content delivery networks



The flexible environment advantages



01

Take advantage of custom configurations and libraries, while focusing on writing code

02

Customize the runtime and the operating system of your virtual machine. Standard runtimes include Python, Java, Go, Node.js, PHP, .NET, and Ruby

03

Customize or provide runtimes by supplying a custom Docker image or Dockerfile

Comparing the App Engine environments

	The standard environment 	The flexible environment 
Instance startup	Seconds	Minutes
SSH access	No	Yes (although not by default)
Write to local disk	No (some runtimes have read and write access to the /tmp directory)	Yes, ephemeral (disk initialized on each VM startup)
Support for 3rd-party binaries	For certain languages	Yes
Network access	Via App Engine services	Yes
Pricing model	After free tier usage, pay per instance class, with automatic shutdown	Pay for resource allocation per hour; no automatic shutdown



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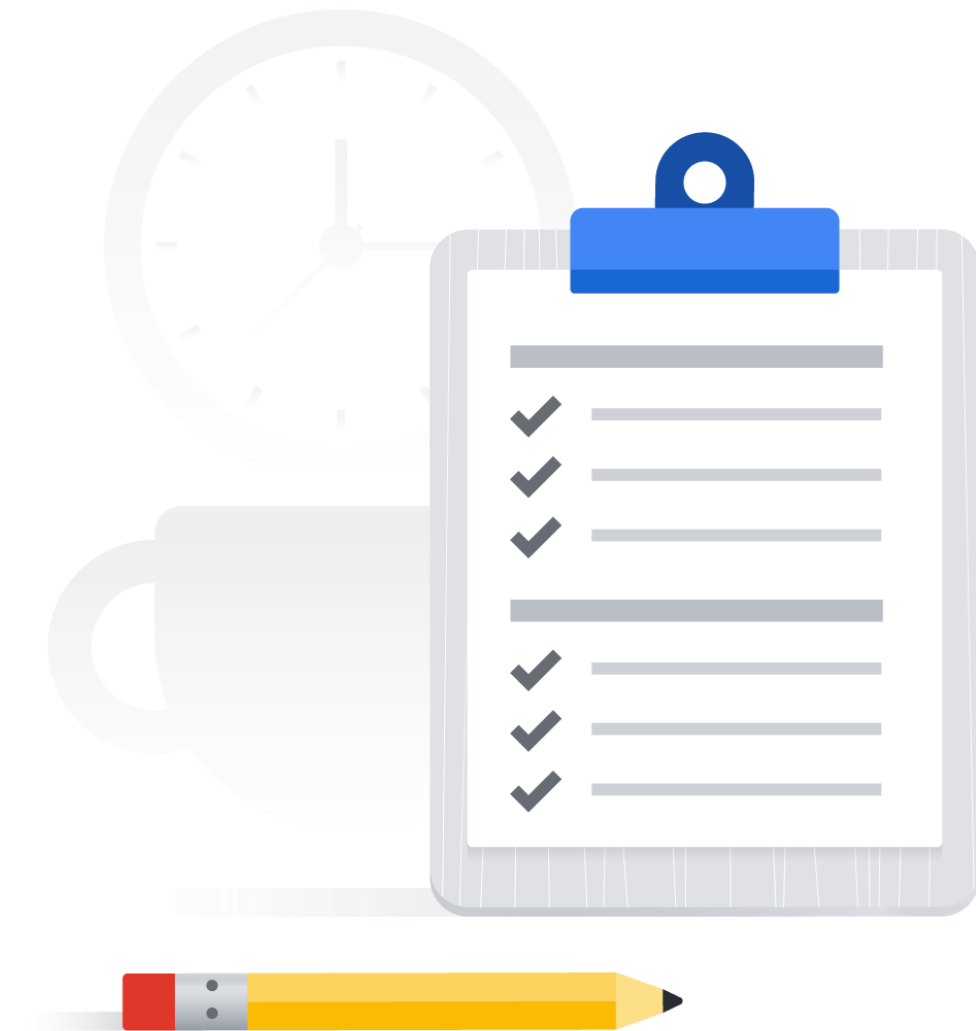
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Lab: App Engine: Qwik Start - Python

 Hands-on lab

During this lab you'll:

1. Download an application.
2. Test the application.
3. Deploy the application.



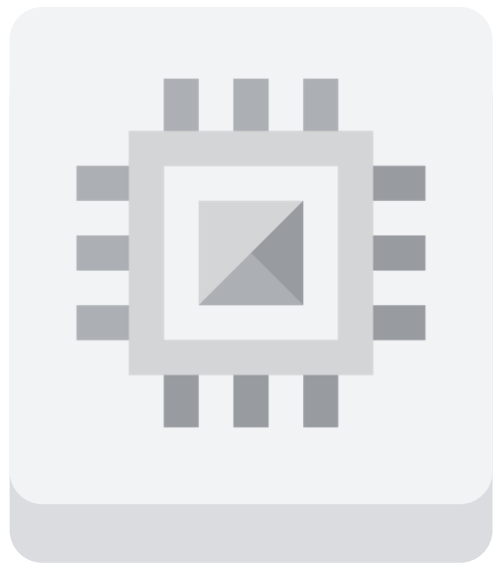


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Google Cloud compute services



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Engine

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App Engine

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Cloud
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Serverless logic

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Google Kubernetes
Engine

Hybrid

Cluster manager and
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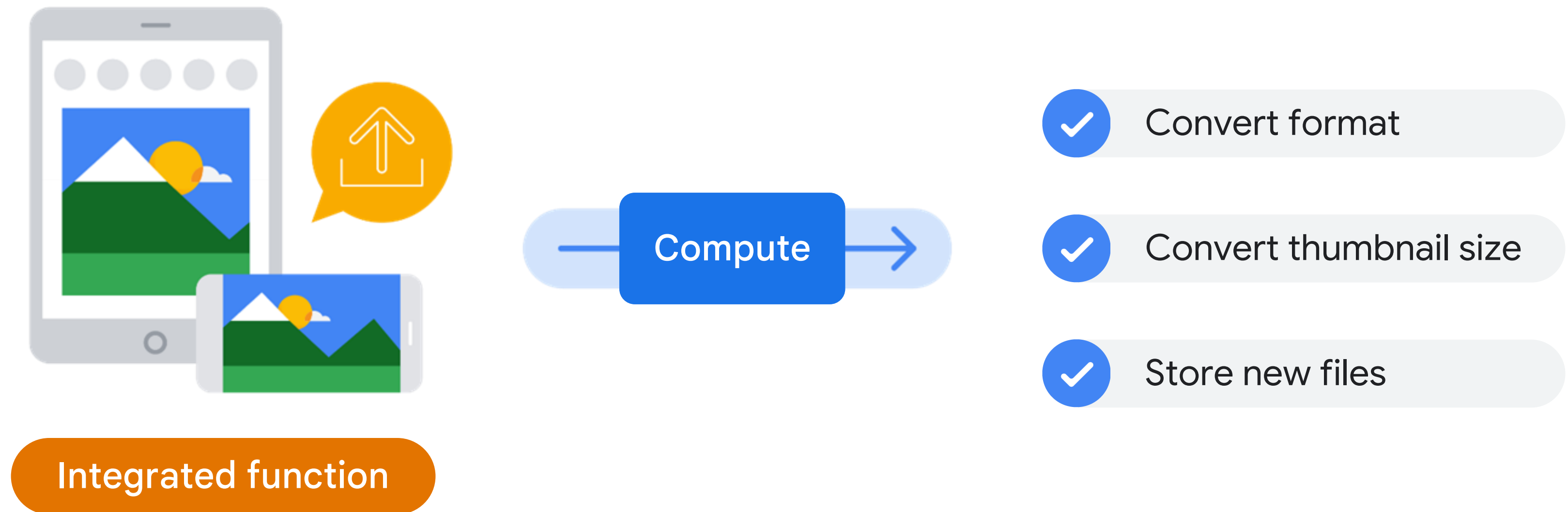


Cloud Run

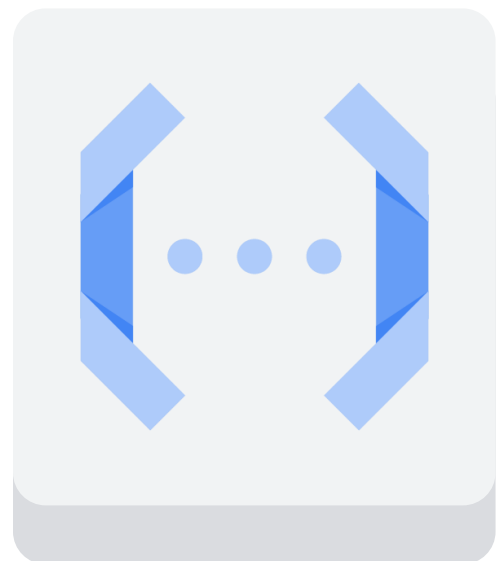
Serverless platform

A fully managed
serverless platform for
developing and
deploying scalable
containerized
applications

Integrated cloud functions handle application events



Cloud Functions allows your code to respond to events



Lightweight, event-based, asynchronous compute solution

Allows you to create small, single-purpose functions that respond to cloud events without the need to manage a server or a runtime environment

Use these functions to construct applications from bite-sized business logic and connect and extend cloud services

Billed to the nearest 100 milliseconds, and only while your code is running

Written in Javascript (Node.js), Python or Go, and execute in a managed Node.js environment on Google Cloud

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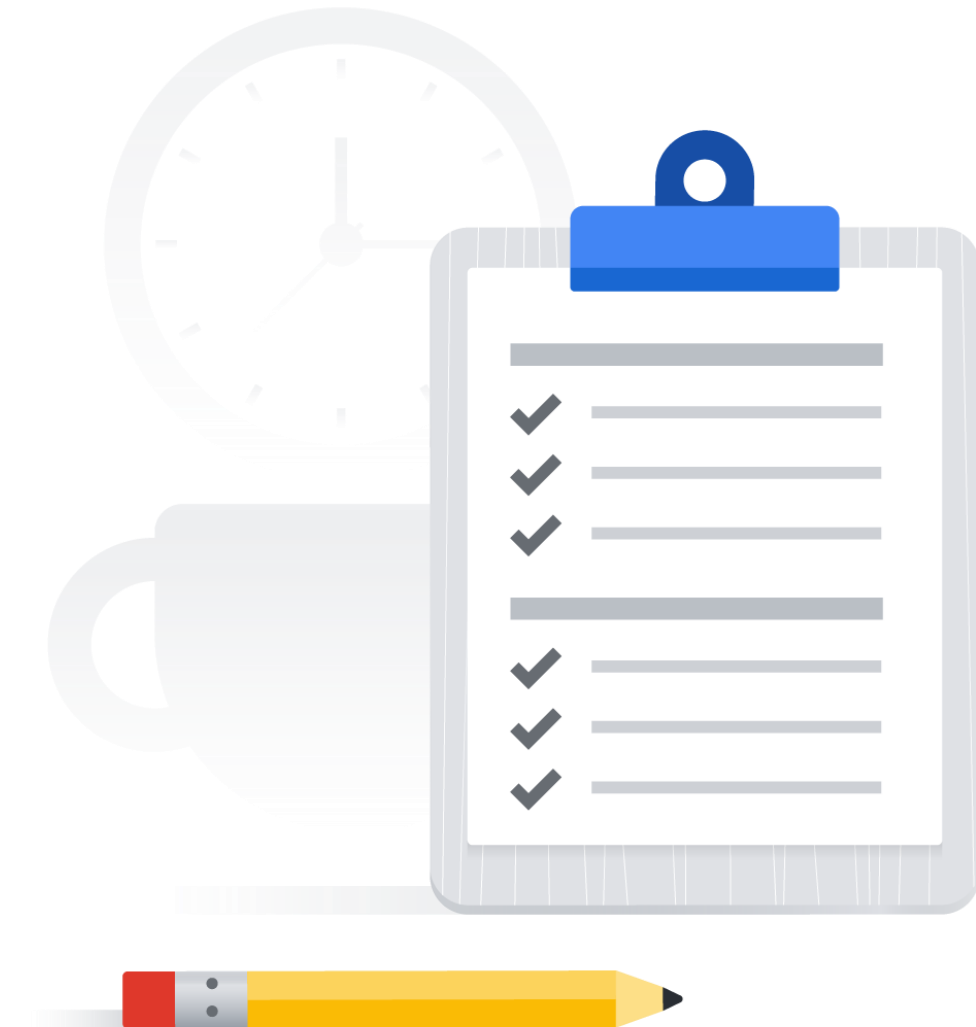
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Lab: Cloud Functions: Qwik Start - Command Line

 Hands-on lab

During this lab you'll:

1. Create a simple cloud function.
2. Deploy and test the function.
3. View logs.



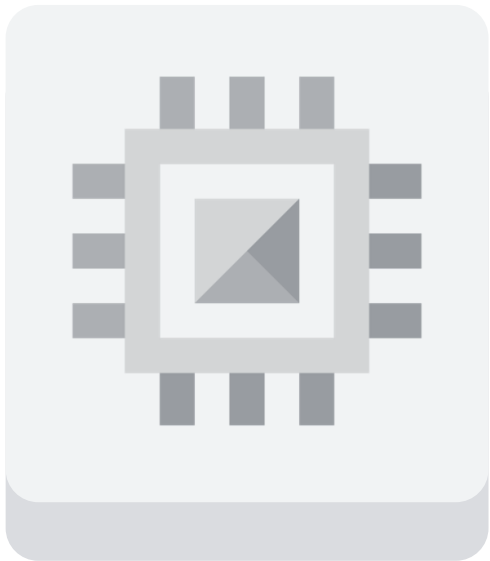


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Google Cloud compute services



Compute
Engine



App Engine



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Google Kubernetes
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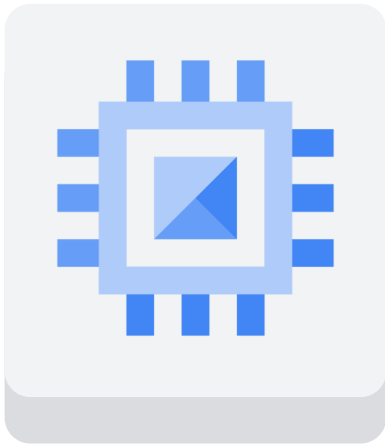
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IaaS to PaaS comparison

Compute Engine

Google
Kubernetes Engine

App Engine



IaaS

PaaS

Servers, networking, storage

Preset run-times, managed services

Pay for what you allocate

Pay for what you use



IaaS

Share compute resources using VMs by virtualizing the hardware using virtual machines



Configurable system

- Customizable installation, configuration and building
- Large, slow and costly



App Engine

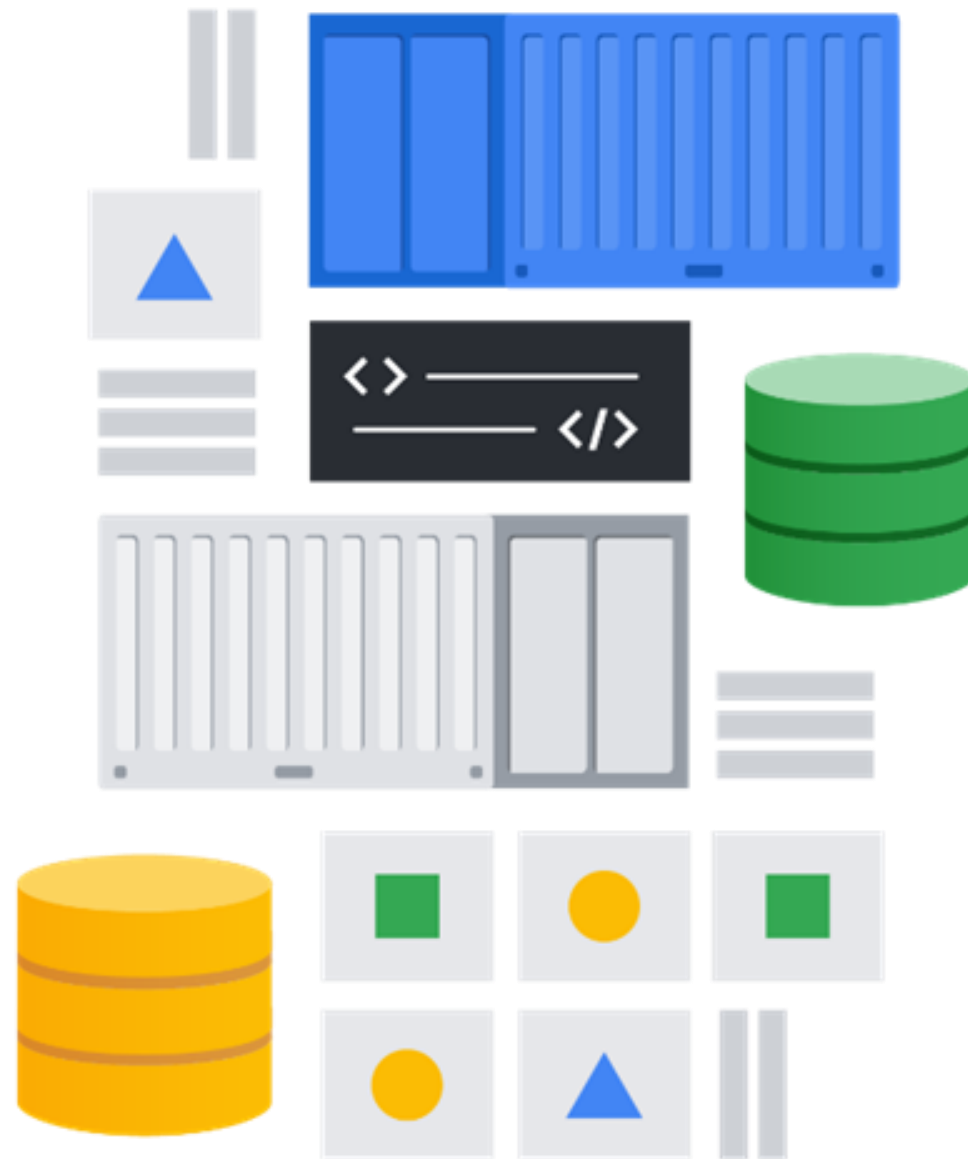
- Access to programming services to write your code
- Seamless, independent and rapid scaling
- No fine-tuning the underlying architecture to save cost



Containers

Independent scalability of workloads in PaaS and an abstraction layer of the OS and hardware in IaaS

Containers group your code and its dependencies



An invisible box around your code and its dependencies

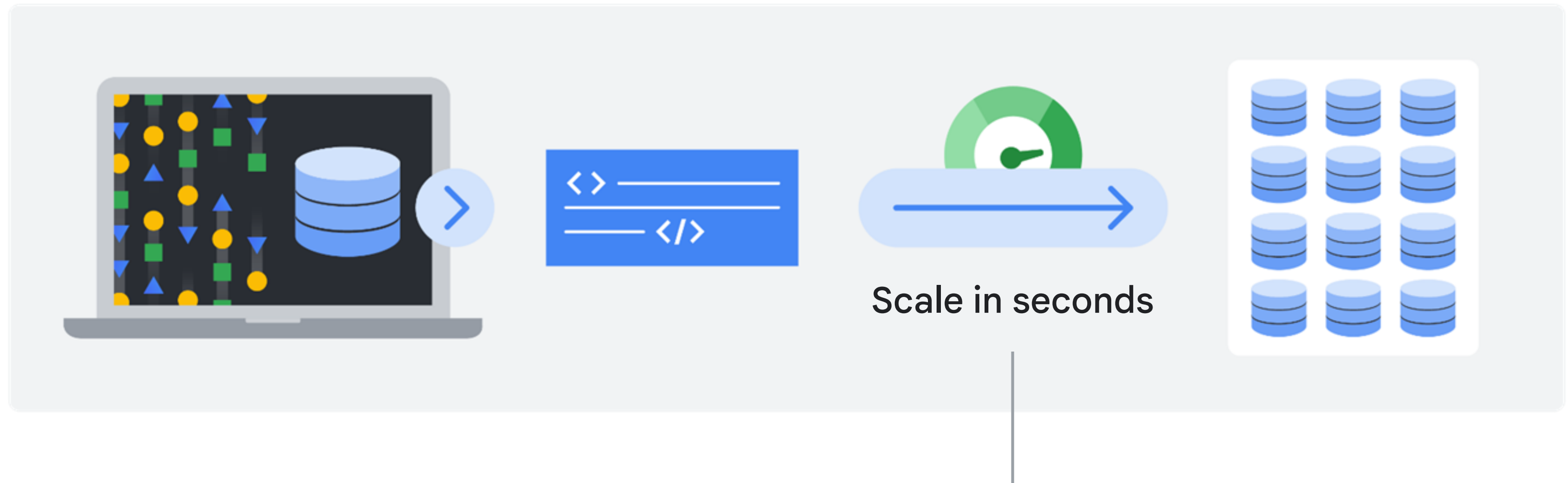
Has limited access to its own partition of the file system and hardware

Only requires a few system calls to create and starts as quickly as a process

Only needs an OS kernel that supports containers and a container runtime, on each host

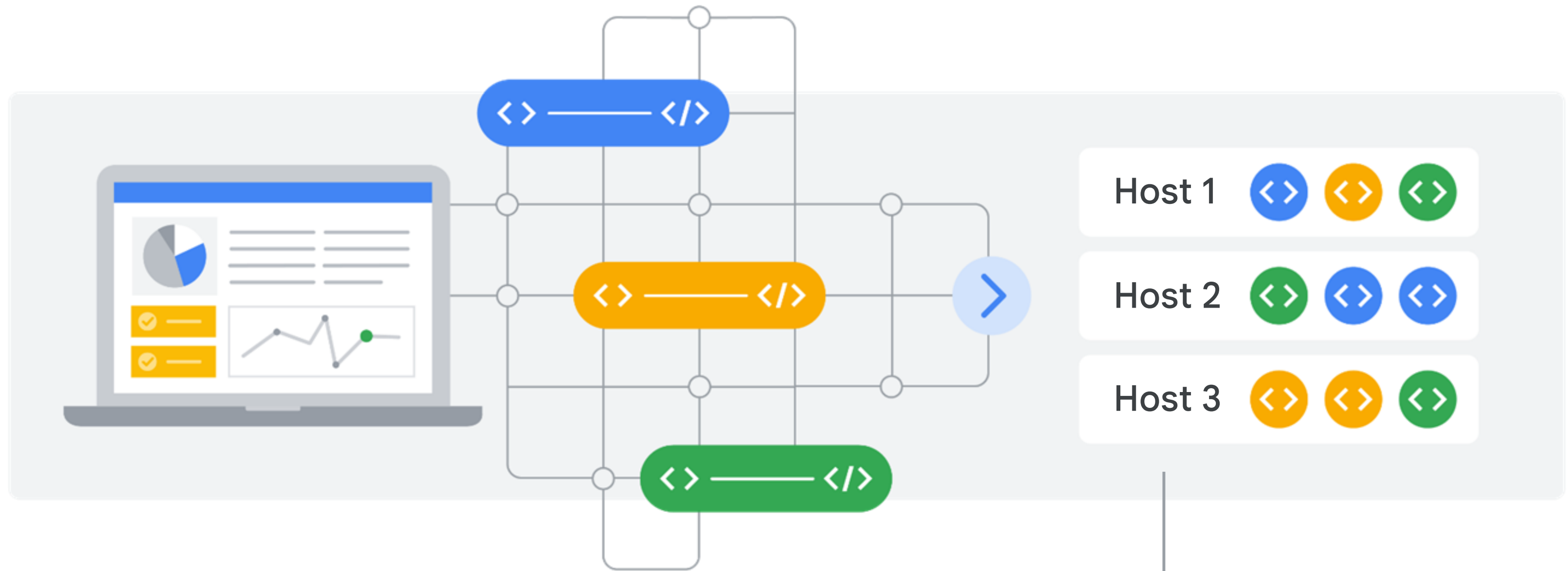
It scales like PaaS, but gives nearly the same flexibility as IaaS

You can scale by duplicating single containers



Scaling one container running the whole web server on a single host

You can also scale an application with multiple containers



Modular, easily deployable, and scale independently across a group of hosts



Kubernetes



A container orchestration tool to simplify the management of containerized environments



Can be installed on a group of your own managed servers or run as a hosted service in Google Cloud



Kubernetes lets you:



Install the system on local servers in the cloud



Manage container networking and data storage



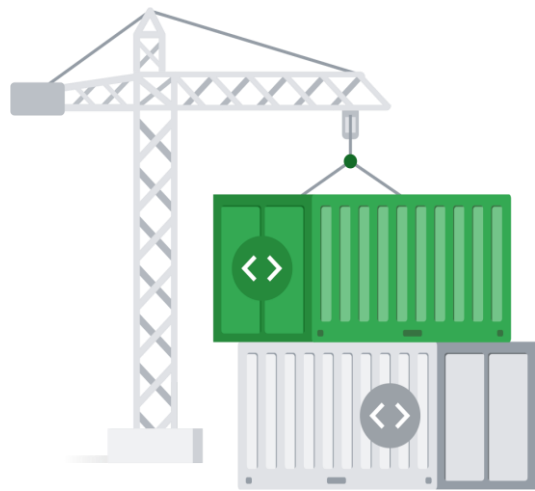
Deploy rollouts and rollbacks



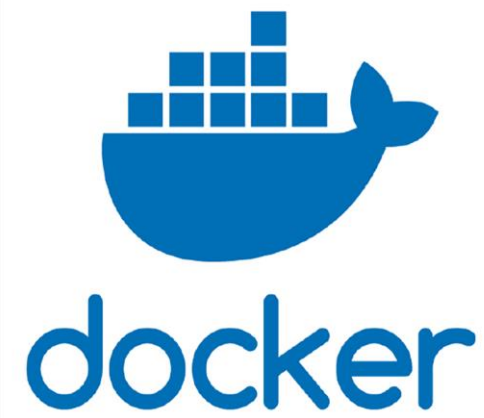
Monitor and manage container and host health

The relationship between containers, Docker, and Kubernetes

Containers



Docker



Kubernetes



Google Kubernetes
Engine (GKE) is a
managed environment
for deploying
containerized apps





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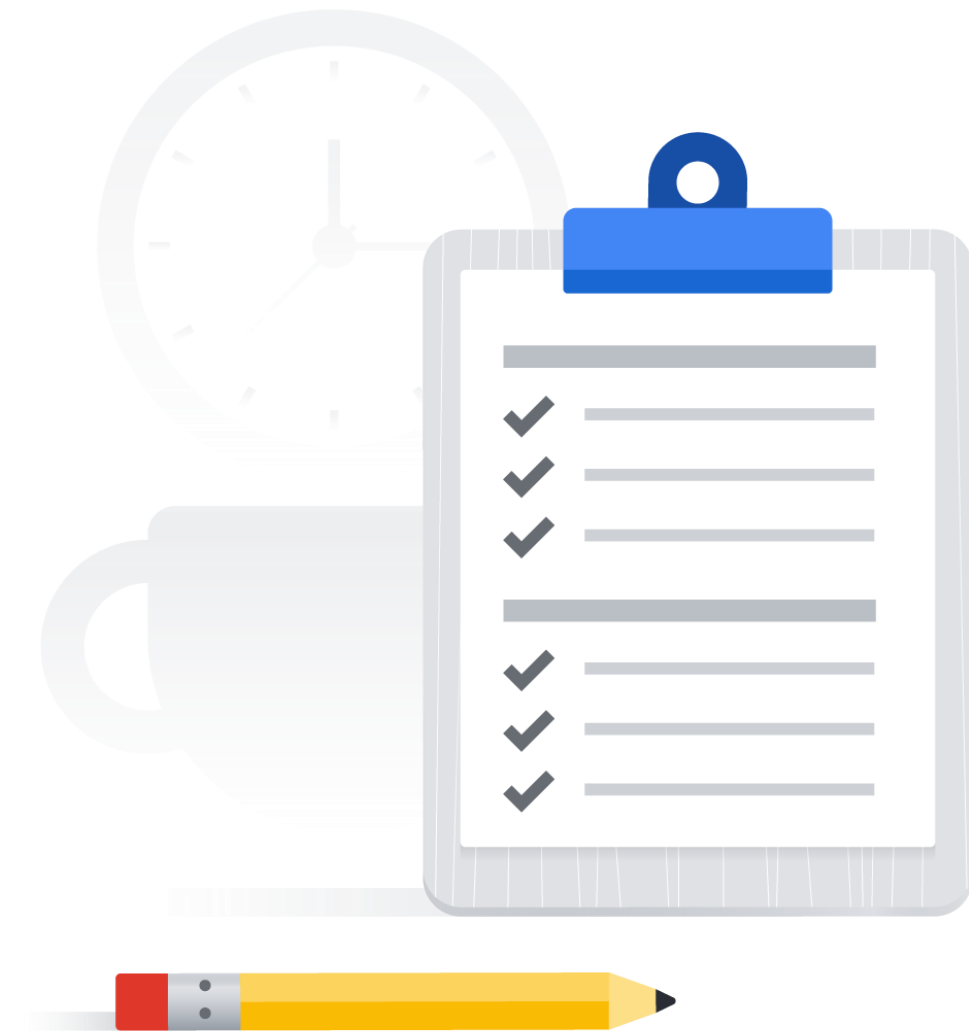
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Lab: Kubernetes Engine: Qwik Start

 Hands-on lab

During this lab you'll:

1. Set a default compute zone.
2. Create a GKE cluster.
3. Get authentication credentials for the cluster.
4. Deploy an application to the cluster.



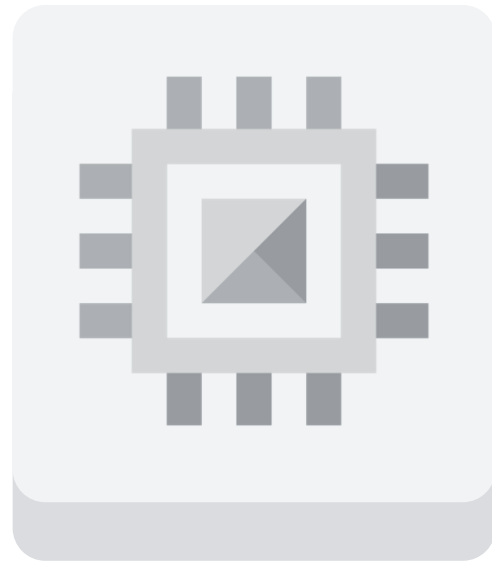


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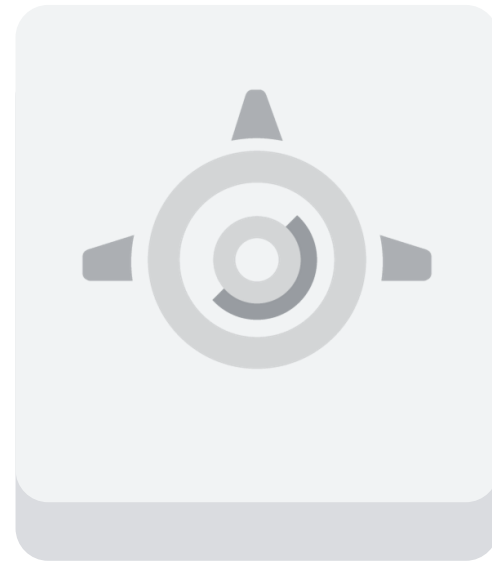


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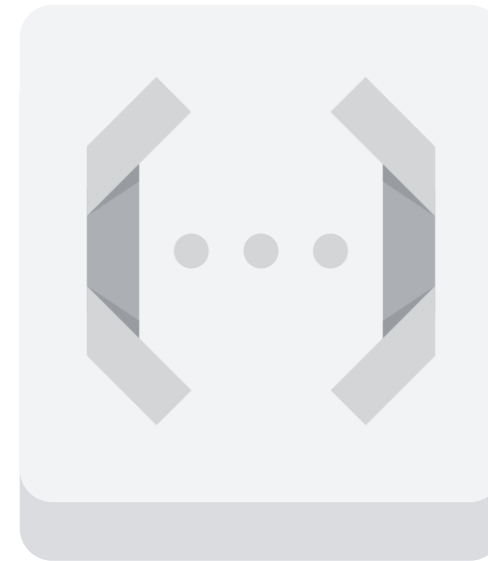
Google Cloud compute services



Compute
Engine



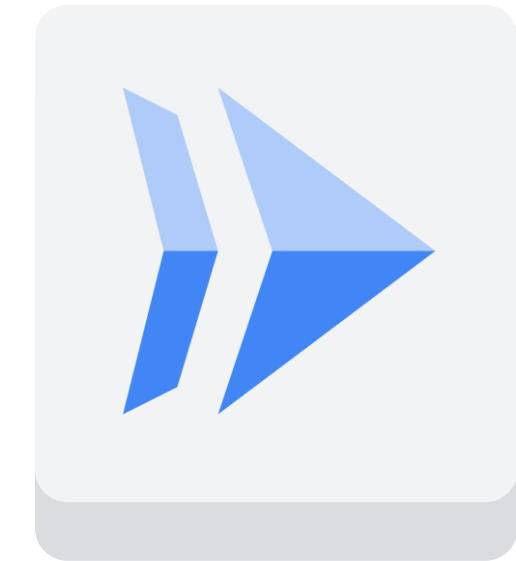
App Engine



Cloud
Functions



Google Kubernetes
Engine



Cloud Run

IaaS

PaaS

Serverless logic

Hybrid

Serverless platform

Virtual machines with
industry-leading
price/performance

A flexible, zero ops
platform for building
highly available apps

A lightweight fully
managed serverless
execution environment
for building and
connecting cloud
services

Cluster manager and
orchestration engine
built on Google's
container experience

A fully managed
serverless platform for
developing and
deploying scalable
containerized
applications

Cloud Run is managed serverless computing



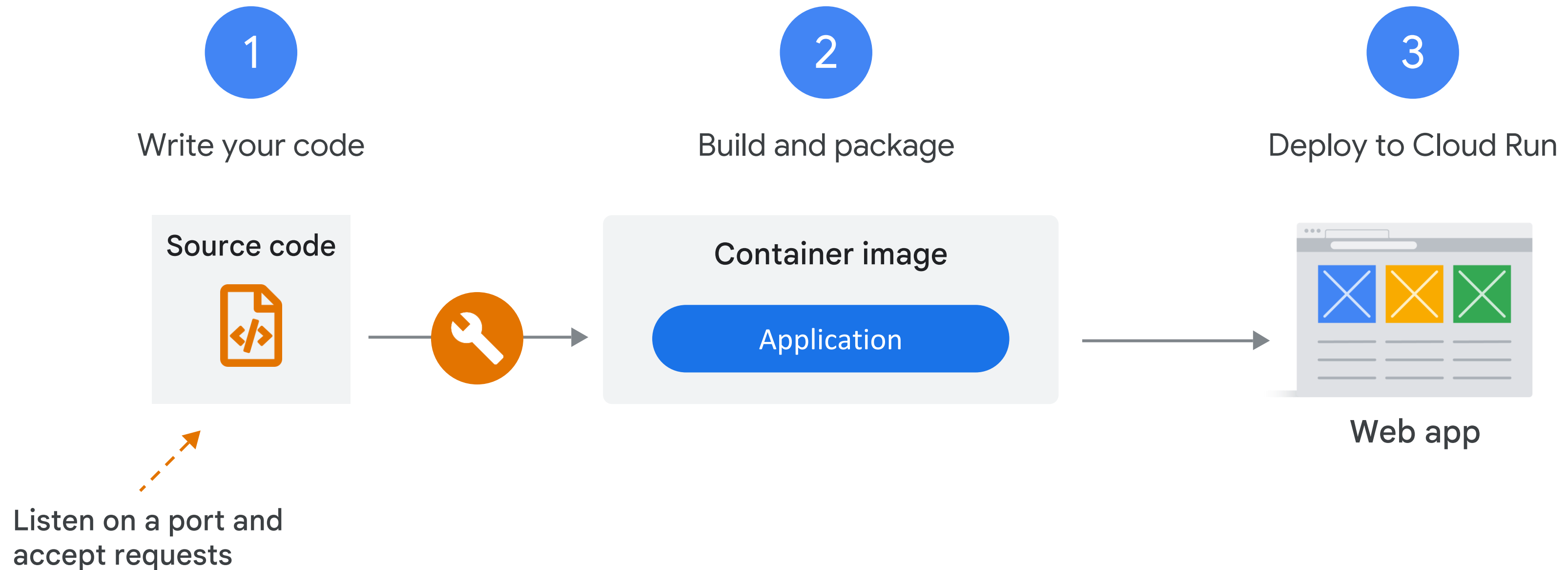
A managed compute platform that can run stateless containers

Serverless, removing the need for infrastructure management

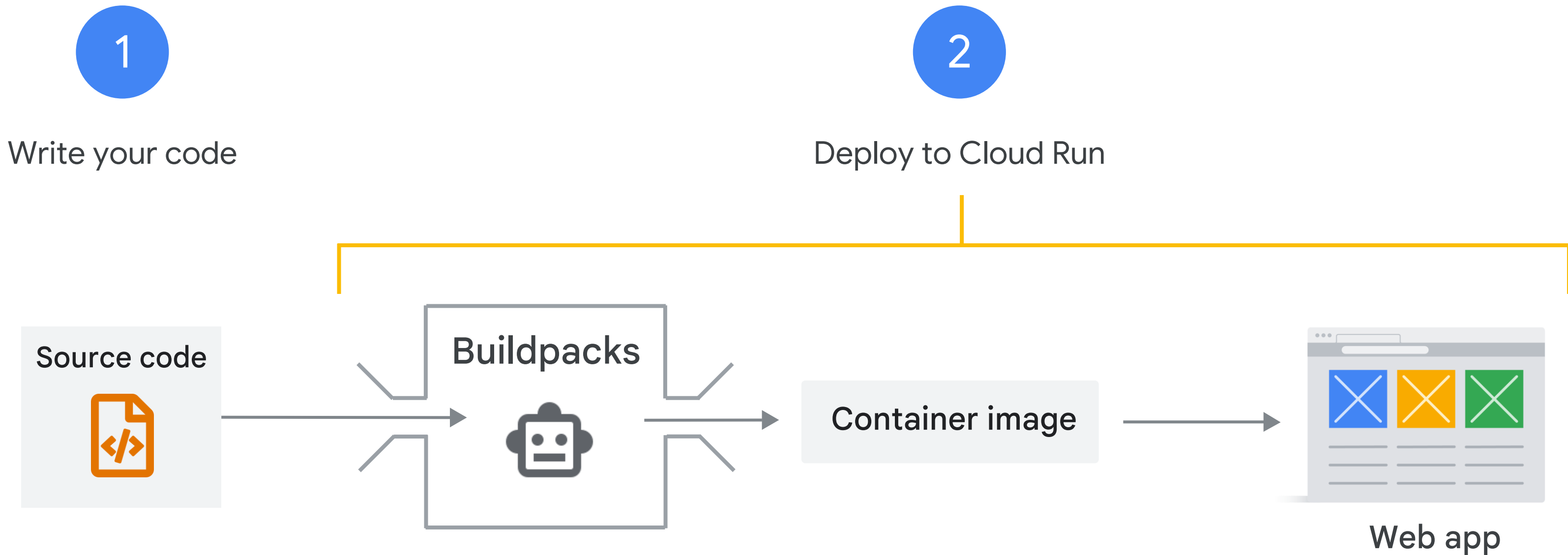
Built on Knative, an open API and runtime environment built on Kubernetes

Can automatically scale up and down from zero almost instantaneously, charging only for the resources used

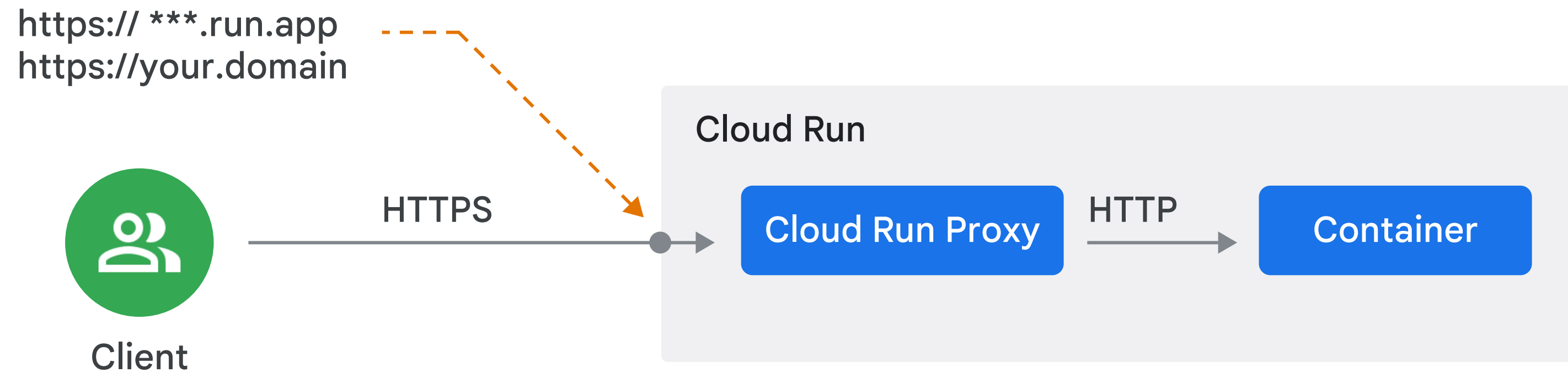
Google Cloud Run workflow is a three-step process



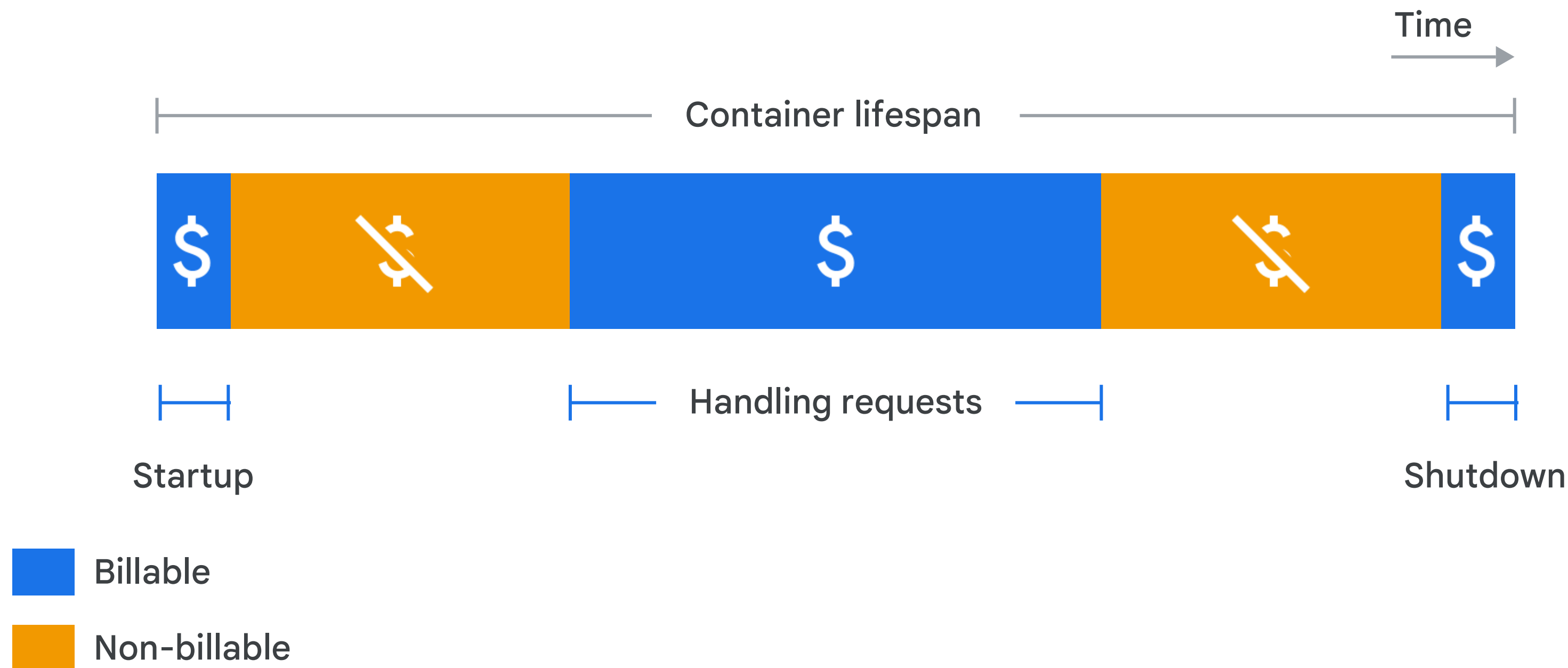
Cloud Run also has a source-based workflow



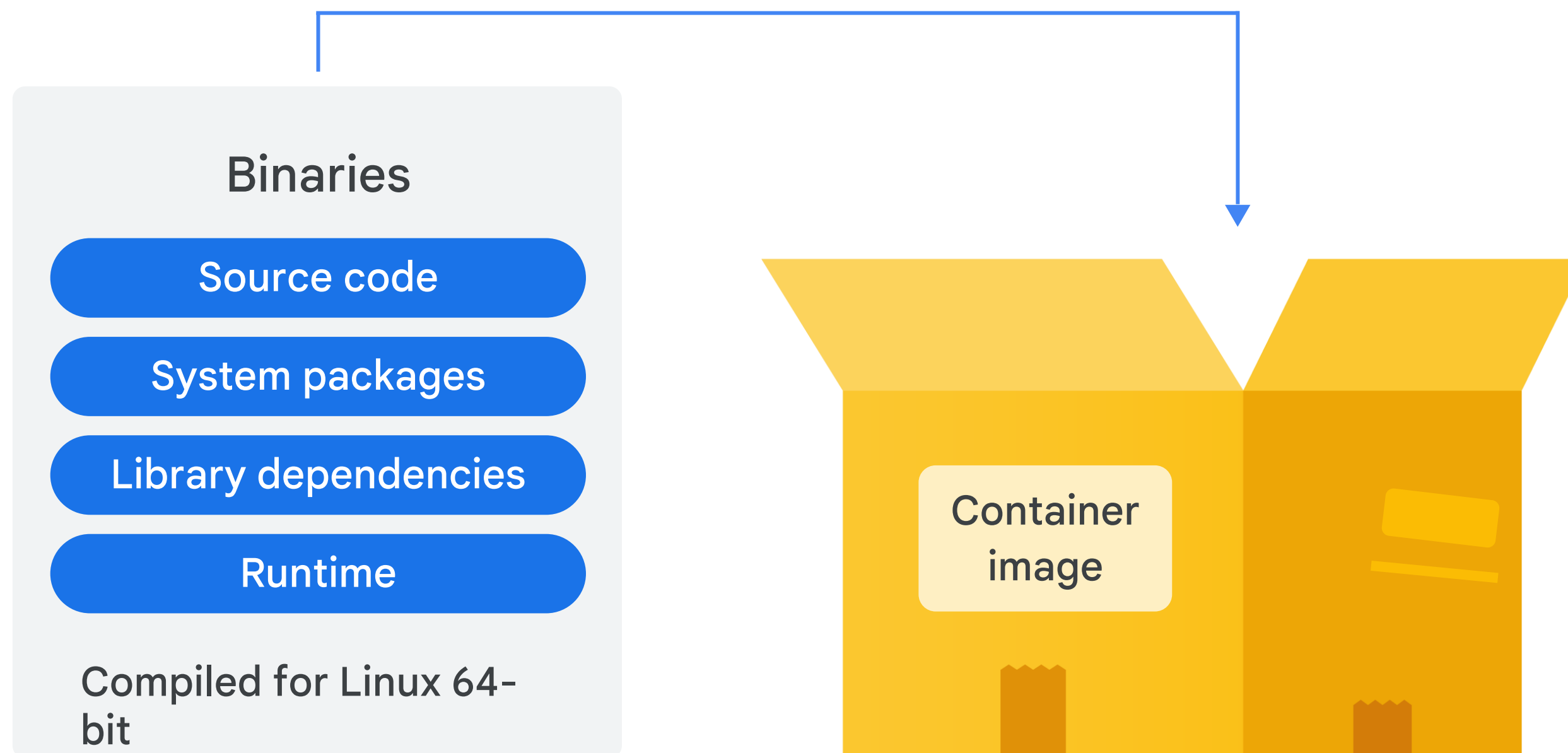
Cloud Run handles HTTPS serving for you



You are only charged when your container handles requests



Cloud Run can run any binary





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Knowledge check

Which compute service would be considered IaaS?

- A. Compute Engine
- B. App Engine
- C. Cloud Functions
- D. Google Kubernetes Engine



Knowledge check

Which compute service would be considered IaaS?

A. Compute Engine

B. App Engine

C. Cloud Functions

D. Google Kubernetes Engine



Knowledge check

What is the Compute Engine feature that allows VMs to be added to or subtracted from an application based on load metrics?

- A. Network time protocol (NTP)
- B. Persistent disks
- C. Autoscaling
- D. Load balancing



Knowledge check

What is the Compute Engine feature that allows VMs to be added to or subtracted from an application based on load metrics?

A. Network time protocol (NTP)

B. Persistent disks

C. Autoscaling

D. Load balancing



Knowledge check

Which App Engine environment is based on preconfigured container instances?

- A. The standard environment
- B. The flexible environment
- C. Both standard and flexible environments



Knowledge check

Which App Engine environment is based on preconfigured container instances?

- A. The standard environment
- B. The flexible environment
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Knowledge check

Which of these is a lightweight, fully managed serverless execution environment for building and connecting cloud services?

- A. Compute Engine
- B. Google Kubernetes Engine
- C. App Engine
- D. Cloud Functions



Knowledge check

Which of these is a lightweight, fully managed serverless execution environment for building and connecting cloud services?

- A. Compute Engine
- B. Google Kubernetes Engine
- C. App Engine
- D. Cloud Functions



Knowledge check

Which of these is a managed compute platform that lets you run stateless containers through web requests or Pub/Sub events?

- A. Cloud Run
- B. Google Kubernetes Engine
- C. App Engine
- D. Cloud Functions



Knowledge check

Which of these is a managed compute platform that lets you run stateless containers through web requests or Pub/Sub events?

A. Cloud Run

B. Google Kubernetes Engine

C. App Engine

D. Cloud Functions



Knowledge check

Which of these is a managed environment for deploying containerized apps?

- A. Cloud Run
- B. Google Kubernetes Engine
- C. App Engine
- D. Cloud Functions



Knowledge check

Which of these is a managed environment for deploying containerized apps?

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B. Google Kubernetes Engine

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Summary

- ✓ Explored the role of compute options in the cloud.
- ✓ Learned about building and managing virtual machines.
- ✓ Examined building elastic applications using autoscaling.
- ✓ Explored PaaS options by leveraging App Engine.
- ✓ Examined building event-driven services using Cloud Functions.
- ✓ Identified containerizing and orchestrating applications with Google Kubernetes Engine.
- ✓ Identified developing and deploying scalable containerized applications with Cloud Run.



